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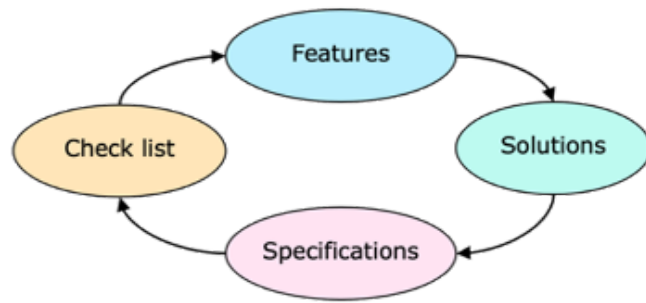
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|--------------|-----------------------------------|
| Started on   | Tuesday, 19 January 2021, 4:00 PM |
| State        | Finished                          |
| Completed on | Tuesday, 19 January 2021, 5:53 PM |
| Time taken   | 1 hour 53 mins                    |
| Grade        | Not yet graded                    |

Question **1**

Complete

Marked out of  
5.00

Exemplify and explain the Usability Design Process presented in the figure for the Visibility and Structure usability principles, in the GUI of a 3D Graphical Editor. Detail each phase and highlight the main issues of such designing approach.



The design process presented in the figure is an iterative one, consisting of 4 phases: Features, Solutions, Specifications and Check list. To create a usability design process, one must go through each of the phases and analyze them.

In our case, the features for which we design are the two usability principles: Visibility and Structure. We will explain these concepts in relation to a 3D graphical editor, such as Blender.

Visibility means that we need to keep all options and materials needed to complete a given task visible to the user. The user must be in control of the execution and be informed of the state of his actions without being distracted with redundant information.

Structure means that the Graphical User Interface must be organized in meaningful and useful ways. The similar controls must be grouped together and the unrelated things must be separated. The models must be consistent and recognizable to users.

Based on these two features, we elaborate a checklist that must be met by our solution:

- controls must be grouped together, based on category
- redundant information is hidden (at least until the user clicks on the related menu)
- inform the user of the state of an operation (successful, unsuccessful, error)
- form validations appear in case malicious input is used

The solutions have been described in the following attachment.

Solution 1 describes an interface with a menu on the bottom, with each category having a separate drop down. The drop down contains related information (such as texture type). On clicking the type, the user sees a prompt with the precision for the texture. The input is validated and an error message appears in case it is out of range. If it is correct, a pop-up appears indicating success to the user.

In Solution 2, We have pictograms and a side menu. On clicking, the user is prompted with a dropdown with the related information (textures) and on clicking them they can adjust a slider for the precision. The success message is displayed in the bottom terminal.

We can see that solution 1 is more clear than solution 2 since it does not have the pictograms, which are ambiguous (for the menu). It also is more intuitive in order to find the desired texture type faster. The solution 2 has the advantage of having sliders.

After choosing the right solution, we will describe it with specifications, and we will check if they complete the checklist.

Specifications for solution 1:

- application has a menu (with words) on the bottom part of the application
- clicking a menu element prompts a drop-down with the details

- success messages appear on performing operations
- forms are validated for range

As it can be seen, the checklist is complete!

 [\\_DesignProcess.pdf](#)

Question **2**

Correct

Mark 1.00 out of 1.00

In which phase of the WebQEM method are aggregated the elementary measurements?

Select one:

- ☒ a. Global evaluation ✓
- ☐ b. Define and specify quality requirements
- ☐ c. Elementary evaluation
- ☐ d. Analysis and documentation of results, conclusions and recommendations

Your answer is correct.

Question **3**

Correct

Mark 1.00 out of 1.00

Rapid prototyping concept could be classified as:

Select one or more:

- ☒ a. Evolutionary ✓
- ☐ b. Parametrical
- ☐ c. Expensive
- ☒ d. Continuous ✓

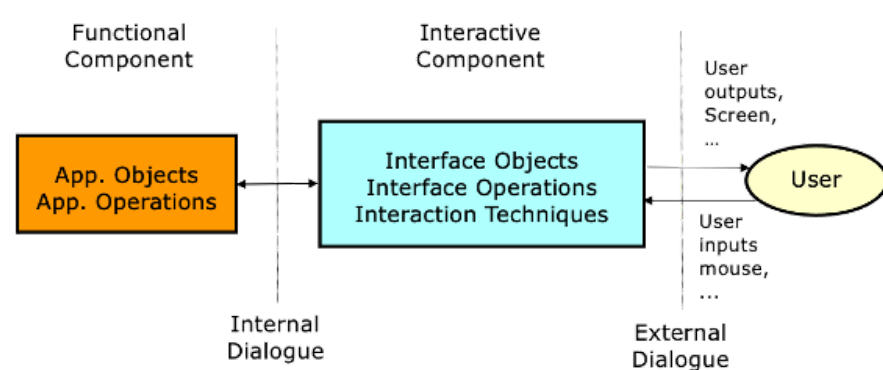
Your answer is correct.

Question **4**

Correct

Mark 1.00 out of 1.00

Dialogue Independence concept concerns with:



Select one or more:

- ☒ a. Support the independent development of interaction techniques and functionality components ✓
- ☒ b. Separate the functional and interactive components ✓
- ☐ c. Support the independent development of interaction techniques and interface operations
- ☐ d. Support the independent development of application objects and application operations

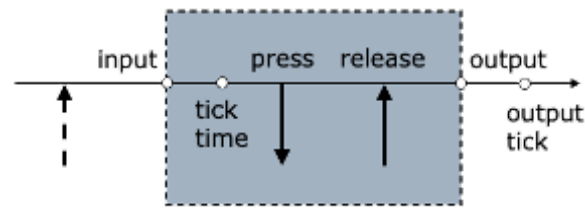
Your answer is correct.

Question 5

Correct

Mark 1.00 out of 1.00

The following figure describes graphically the following gesture as a user input technique:



Select one:

- ☐ a. Range gesture
- ☒ b. Press-timer gesture ✓
- ☐ c. Click gesture
- ☐ d. Drag gesture

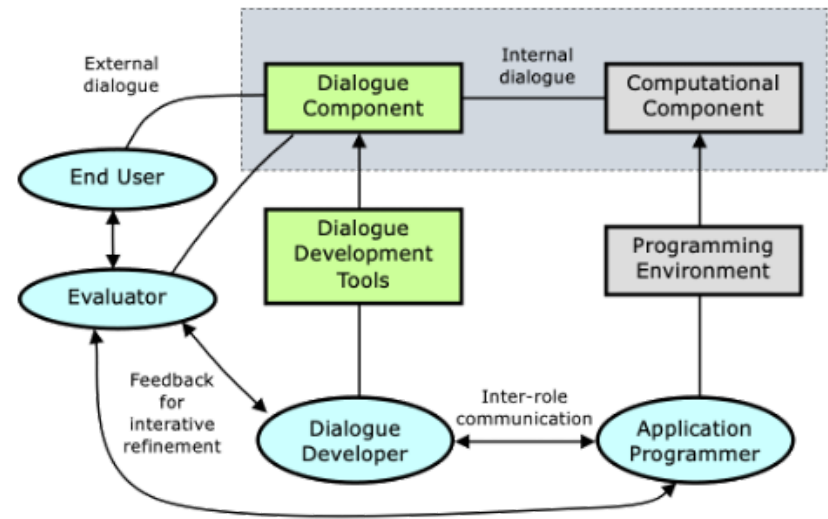
Your answer is correct.

Question 6

Correct

Mark 1.00 out of 1.00

The following schema describes:



Select one or more:

- ☒ a. Interactive software development methodology ✓
- ☐ b. Interactive software application consists of Dialogue Component and Programming Environment
- ☒ c. Dialogue Independence concept used in software development methodology ✓
- ☐ d. Interactive software application consists of Dialogue Development Tools and Computational Component

Your answer is correct.



Question **10**

Complete

Marked out of  
5.00

Exemplify and explain the extension of some usability heuristics in order to discover and identify detailed usability problems. Exemplify the heuristic: Match Between System and Real World, for application GUI: controlling the movement of an avatar within a video game.

The Match between system and the Real World heuristic refers to the fact that real world conventions and real world user language are used in the GUI. The system should speak the user's language with concepts familiar to the user rather than technical terms. Information should appear in a natural and logical order.

In the case of the movement of an avatar within a video game, we add supplementary explanations to the heuristic in order to help the people evaluating the game focus on the important elements. The person who evaluates will need to check each of the questions and give a mark (percentage) for the heuristic, in order to identify flaws and possible solutions.

The questions to extend the heuristic would be:

- Does the avatar move in a manner similar to the mental model that the user has from real life?
- Is it similar to other games?
- Are the notions (directions, movement patterns, jumps etc.) familiar to the user?
- Is the information displayed naturally?
- Is the movement realistic? Does the avatar look human-like?
- Are there any uncertainties?

In the case of the game, the user is represented by a partial (hand, etc.) or a full body avatar. It can be represented as first or third person (see from above or see on his own). His presence may be controlled by a number of keys (up, down, left, right or W,A,S,D), the mouse (clicks and movements), joysticks, or VR kits. The user's presence can cause problems if the controls are not chosen correctly. The navigation of the avatar must be intuitive, such as

Up/W key/Left click : go forward

Down/S key/Right click: go back

Left/A key/Move with mouse and go forward: go left

Right/D key/Move with mouse and go forward: go right

Space: jump

Scroll: see more or less

In case the user is given different controls, since they may be accustomed from other games, they will move incorrectly.

If the avatar moves unnaturally, the movement will not be similar to the real world.

If there are different notions used for movement (such as move avatar to forward by fast), the user may be confused.



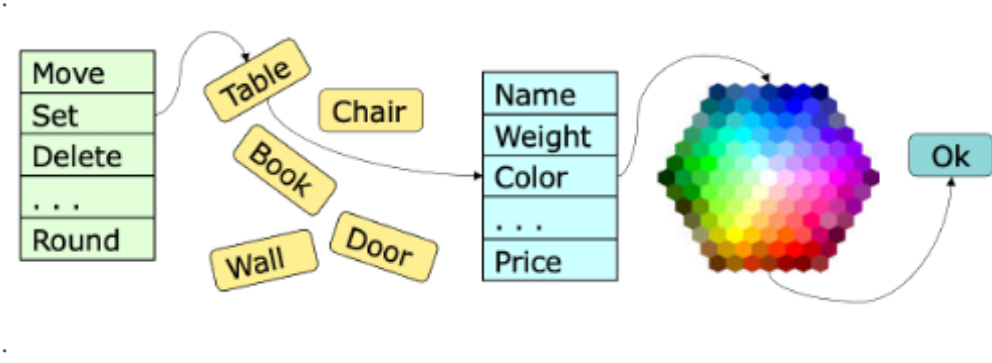
Question **11**

Complete

Marked out of 5.00

Exemplify and explain the using of the concept of Dialogue Cell in the graphical interface in the figure, to execute the following command: Set Table Color to RED <ok>

Highlight and detail the issues related with the elements of the dialogue cell.



Dialogue cells are effective UI components that help users achieve their goals faster and easier. A dialog is an pop-up/overlay which requires user interaction. They are similar to an interactive form, requesting users to make decisions and involving them in tasks. They direct user's attention to a specific task without taking away their context of the current screen. A dialog should be kept simple and speak the user's language.

For the example of "Set table color to RED" <ok>, the user could be presented with the attached interface:

They have a table. On clicking the table, they have a settings wheel pictogram which, on click, opens the Settings Dialogue cell. The current context is still the context of the table.

The settings dialogue cell contains only the table related information, such as set color, set width, change columns, etc. It is short and concise.

The user can select the set color operation from a dropdown menu. Until then, the color picker is hidden and the OK button disabled.

The color picker then appears and the user selects the red color from the picker.

After selecting the color, the OK button is enabled and the user can complete the tasks. The dialogue will be closed and the user will be returned to the previous context.

- Issues
- dialogue must not be crowded with information
  - on pressing ok, the user cannot see the operation anymore
  - the color picker is hidden until user presses
  - it may be difficult for the user to hint how to get to the point where the dialogue opens
  - the options must be clear and concise
  - dialogue should provide feedback that the operation is complete

 [Dialogue-cell.pdf](#)

Question **12**

Correct

Mark 1.00 out of 1.00

The Output Communication model could be:

- Select one or more:
- ☒ a. Continue refresh model ✓
  - ☒ b. Message model ✓
  - ☐ c. Linguistic model
  - ☐ d. Graphical model

Your answer is correct.

Question **13**

Correct

Mark 1.00 out of 1.00

What data gathering technique could be directed to a consensus view and could highlight areas of conflict?

Select one:

- ☐ a. Individual interview
- ☐ b. Naturalistic observation
- ☒ c. Group interview ✓
- ☐ d. Questionnaires

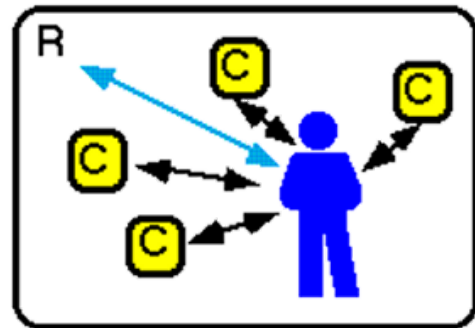
Your answer is correct.

Question **14**

Correct

Mark 1.00 out of 1.00

The following figure describe the concept of:



Select one:

- ☐ a. Virtual Reality User Interface (VRUI)
- ☐ b. Graphical User Interface (GUI)
- ☒ c. Ubiquitous Computer User Interface (UCUI) ✓
- ☐ d. Augmented Reality User Interface (ARUI)

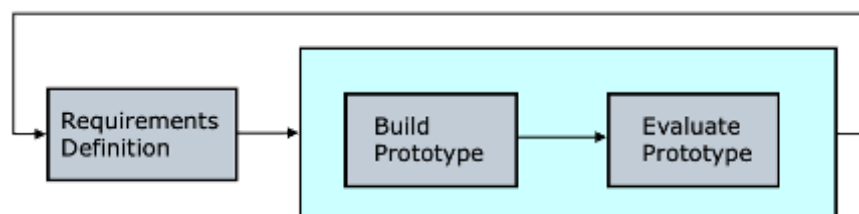
Your answer is correct.

Question **15**

Correct

Mark 1.00 out of 1.00

The figure describes the prototyping approach by:



Select one:

- ☒ a. Evolutionary prototyping ✓
- ☐ b. Vertical prototyping
- ☐ c. Incremental prototyping
- ☐ d. Revolutionary prototyping

Your answer is correct.

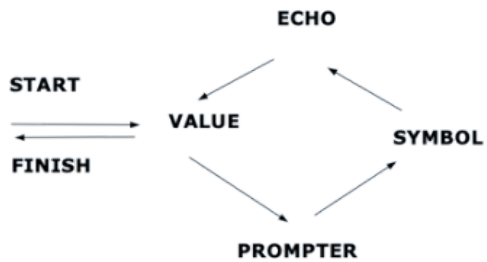


Question **16**

Correct

Mark 1.00 out of 1.00

The following graph represents the user interaction cycle through:



- Select one or more:
- ☒ a. Push button ✓
  - ☐ b. Monitor
  - ☐ c. Mouse
  - ☒ d. Edit box ✓

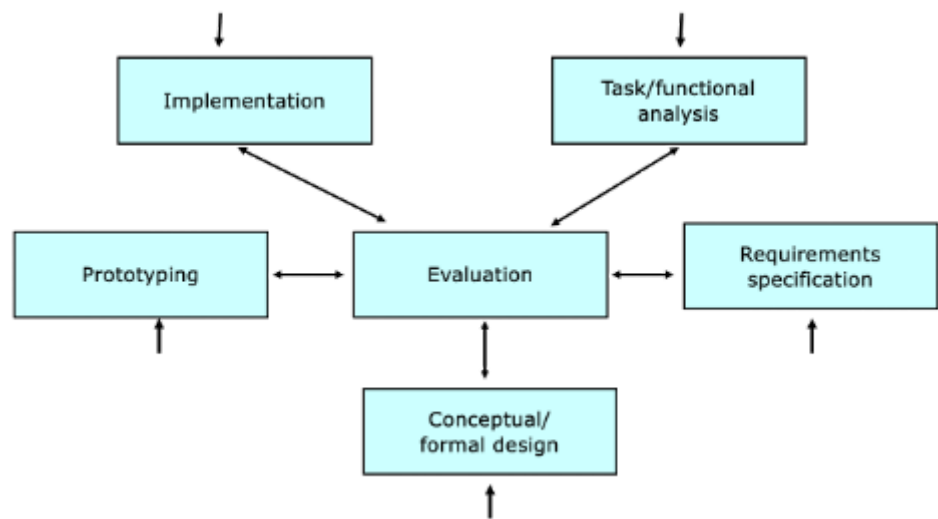
Your answer is correct.

Question **17**

Correct

Mark 1.00 out of 1.00

The following Software Development Methodology has the following features:



- Select one or more:
- ☐ a. Require consistent formal definition of the tasks
  - ☒ b. Appropriate for Web Site development projects ✓
  - ☒ c. Stop and resume of the development could be on any phase ✓
  - ☐ d. Strict sequence of phases

Your answer is correct.

Question **18**

Incorrect

Mark 0.00 out of 1.00

Fitts's Law based evaluation is performed in the following phase of the software development methodology:

- Select one:
- ☐ a. Analysis phases
  - ☒ b. Final test and evaluation phase ✗
  - ☐ c. Implementation phase
  - ☐ d. Design phase

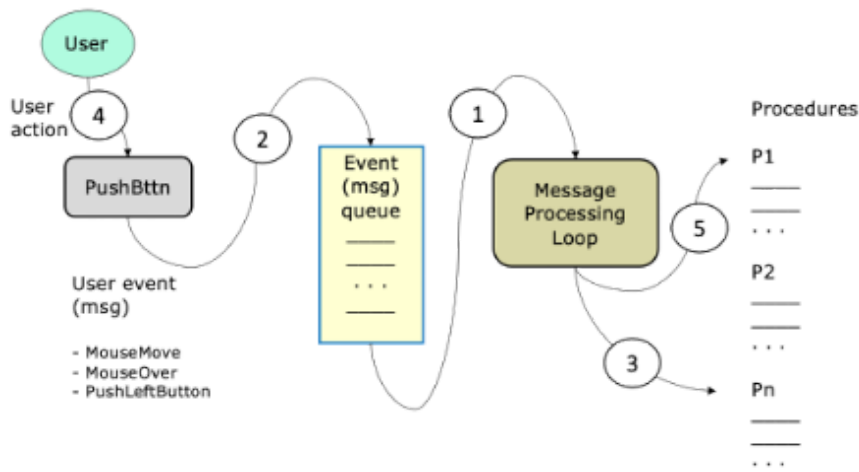
Your answer is incorrect.

Question **19**

Correct

Mark 1.00 out of 1.00

The event oriented flow control sequence throughout the following architecture is:



Select one:

- ☐ a. 1, 2, 3, 4, 5
- ☐ b. 4, 2, 5, 1, 3
- ☒ c. 4, 2, 1, 5, 3 ✓
- ☐ d. 3, 5, 1, 4, 2

Your answer is correct.

Question **20**

Incorrect

Mark 0.00 out of 1.00

What user communication channel concerns with the all 4 following levels of the interaction language: conceptual, semnatic, syntactic, lexical?

Select one or more:

- ☒ a. Graphical based communication language ✗
- ☐ b. Direct manipulation based communication language
- ☐ c. Voice based communication language
- ☐ d. Linguistic based communication language

Your answer is incorrect.

Question **21**

Correct

Mark 1.00 out of 1.00

The following interactive technique is a gesture:

Select one or more:

- ☐ a. menu
- ☒ b. press-timer ✓
- ☐ c. radio button
- ☒ d. press-down ✓

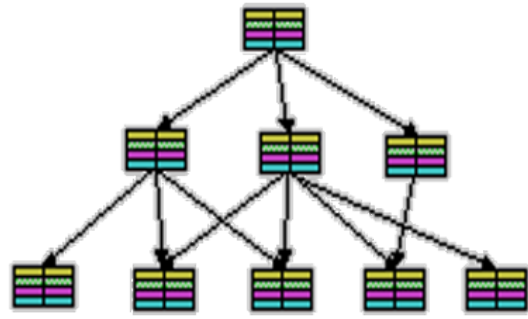
Your answer is correct.

Question **22**

Correct

Mark 1.00 out of 1.00

The following figure describes graphically the following menu as a complex user input interaction technique:



Select one:

- ☒ a. Acyclic network menu ✓
- ☐ b. Linear sequence menu
- ☐ c. Cyclic network menu
- ☐ d. Tree structure menu

Your answer is correct.

Question **23**

Complete

Marked out of 5.00

Exemplify and explain the role of the Recognisers and Commands modules in the Graphical User Interfaces. Give examples of 3D Graphics editing.

Recognizers and commands have the role to gather input.

The attached figure explains how they work: an event happens in the Interactive Component, triggered by the user, using input communication concepts.

For example, the user can click a menu and input some information into a form. Then, Recognizers have the purpose of deciphering the user event and attribute it to a certain command, which the system will understand.

The Command module stores predefined commands such as SET, CHANGE, ADD, SAVE, etc. which are further on associated with an operation, which the functional component can fulfil.

Therefore, the functional component is "fed" with commands which come from the interactive component.

So, recognizers and commands are part of the interactive component and they

- check for input correctness
- decode the user inputs
- gather inputs towards forming an operation
- transform the input into an available form to the operations
- send the inputs to the operations in the functional component

With respect to 3D Graphics editing, the UI must gather information for a user event: such as Zoom in to the cube with 15%. The recognizers then recognize which the action is (zoom in) and which its parameters are (cube, 15%).

The command to which it will associate the action will be ZOOM, which will be then sent to the functional component.

The functional component will then execute the operation.

 [Recognizers-And-commands.pdf](#)

Question **24**

Correct

Mark 1.00 out of  
1.00

The Formative Evaluation of the usability that expect to find out answers to the following questions is:

- Does this action support the completion of the proposed task?
- Will thuser notice that the action is available?
- Will the user understand the feedback given by the action?

Select one:

- ☐ a. Review based evaluation
- ☒ b. Cognitive walkthrough ✓
- ☐ c. Model based evaluation
- ☐ d. Heuristic evaluation

Your answer is correct.

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Interface Design ▶](#)